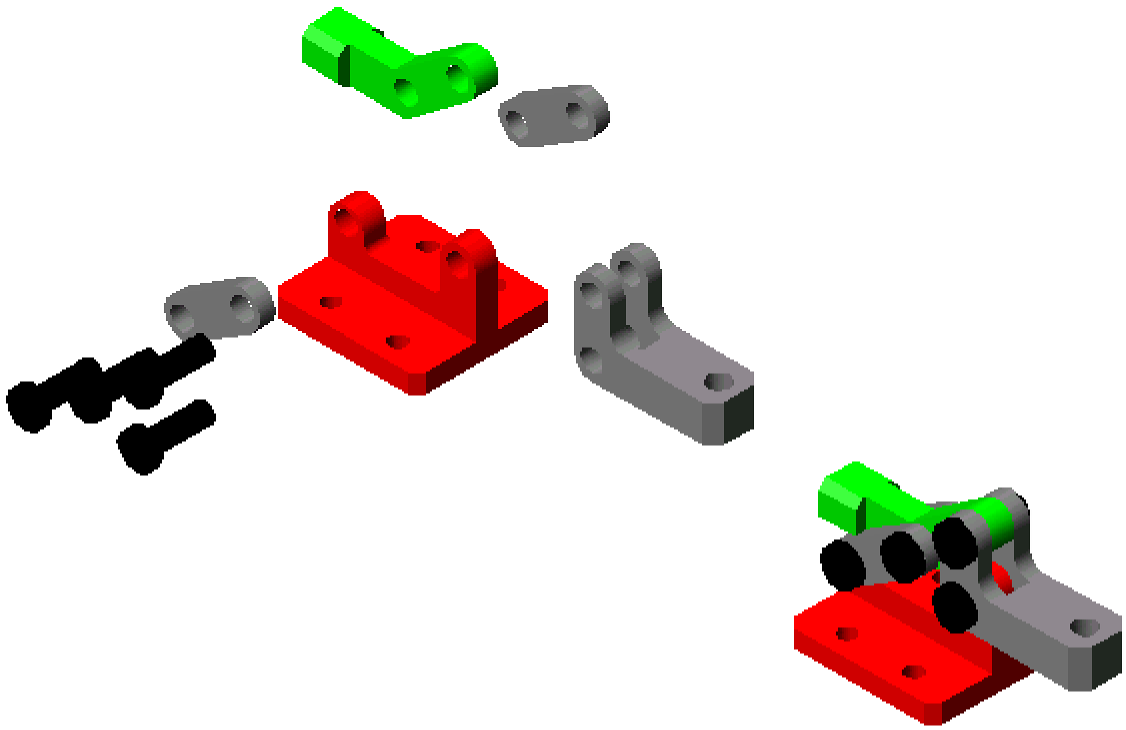
Basic Machine Elements



Basic Machine Elements

Shafts

Couplings

Gears

Flexible Power-Transmitting Elements

Threaded Fasteners

Roller Bearings

Others

A **shaft is** a rotating **machine** element, usually circular in cross section, which **is used** to transmit power from one part to another. The various members such as pulleys and gears **are** mounted on it.

Basic Machine Elements

DEFINITION TYPES IDENTIFICATIO

Couplings
Gears
Flexible Power-Transmitting Elements
Threaded Fasteners
Roller Bearings
Others

Shaft **couplings are used** for power and torque transmission between two rotating shafts.

Couplings

Gears

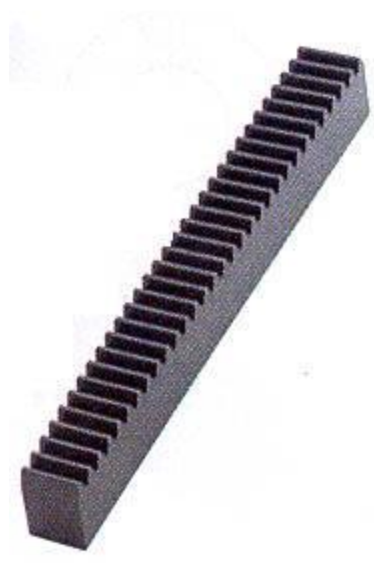
Flexible Power-Transmitting Elements

Threaded Fasteners

Roller Bearings

Others

A toothed mechanism used to transmit motion either to shafts (parallel or perpendicular) or racks.

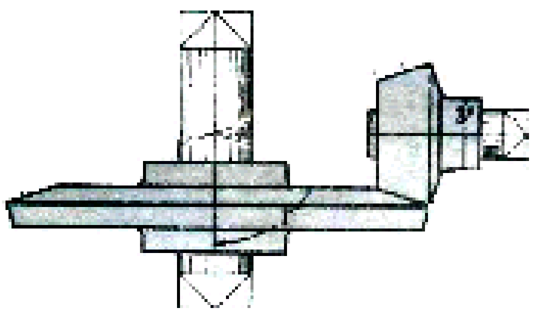


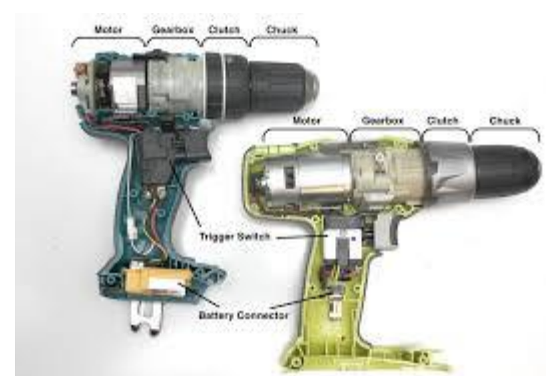
Basic Machine Elements

Spur **Gears** are usually **used** for "slow speeds."

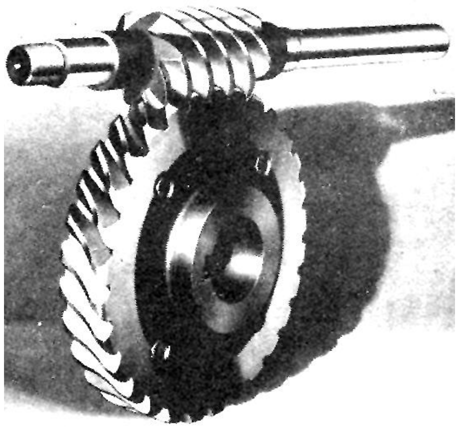
Basic Machine Elements

Bevel **Gears are used** for objects that reach higher speeds.





Worm **Gears** are often **used** to lock the **gears**.



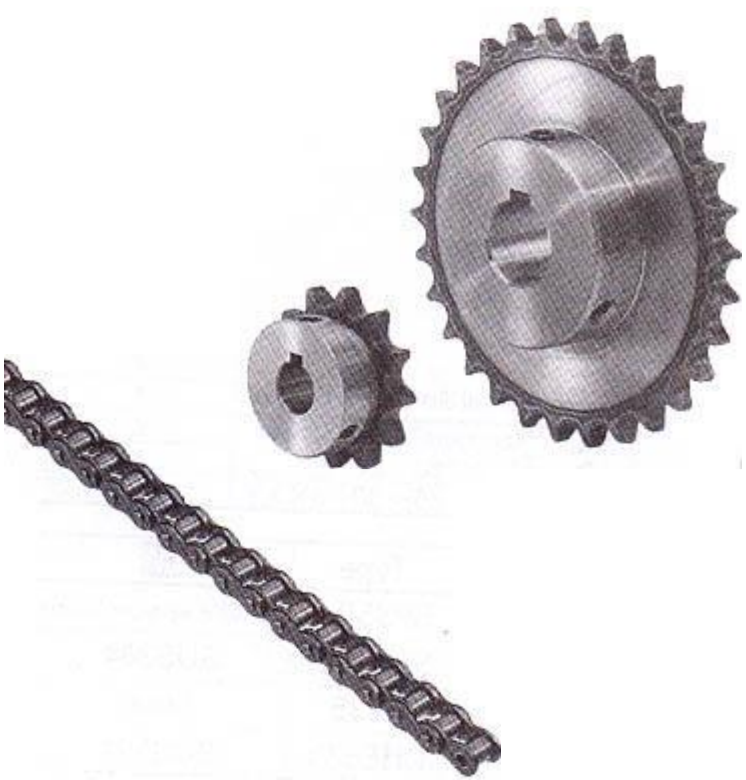
Basic Machine Elements

Flexible drives are generally used for the transmission of power over comparatively long-distances.

Flat belts are forgiving of misalignment; however, proper alignment improves **belt** life for any drive.

Usually made of fabric and cord and impregnated with rubber. **V-Belt drive** system, the thread is the **V-Belt** and the rollers are called as pulleys. **V-Belts** are friction based power or torque transmitters. The power is transmitted from one pulley to the other **by** means of the friction between the **belt** and pulley.

Roller **chain** and sprockets is a very efficient method of power transmission compared to (friction-drive) **belts**, **with** far less frictional loss.



Basic Machine Elements

DEFINITION **TYPES** **IDENTIFICATIO** **ADVANTAGES**

Couplings

Gears

Flexible Power-Transmitting Elements

- Flat Belt
- V-Belt
- Chain
- **Timing Belt**

Threaded Fasteners

Roller Bearings

Others

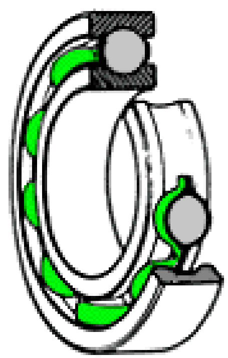
- High Mechanical Efficiency
- Non-slip operation (positive drive)
- Low maintenance cost
- Quiet operation
- High initial tension not necessary
- Large capacity in small space
- Can tolerate low arc of contact (minimum of 6 teeth in contact for rated power)
- Minimal backlash

- Belts are selected according to the width of the belt
- Belts must be selected from the variable standard belt widths

- A **threaded fastener** is a discrete piece of hardware that has internal or external screw threads. It falls into the overall fastener family. They are usually used for the assembly of multiple parts and facilitate disassembly.

Basic Machine Elements

A **bearing** is a machine element that constrains relative motion to only the desired motion, and reduces friction between moving parts.



Couplings

Gears

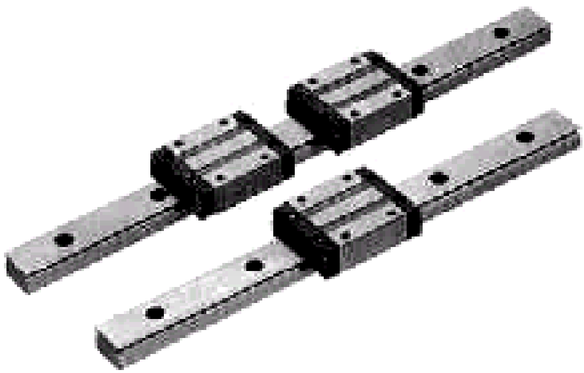
Flexible Power-Transmitting Elements

Threaded Fasteners

Roller Bearings

Others

- **LM Rail:** used as guides for sliding systems hence, also called as *Slide Guides*

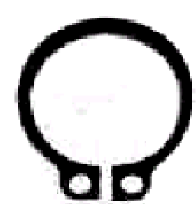


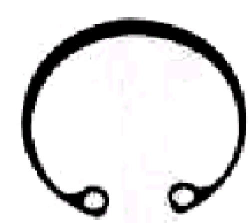
- **Ball Screw:** used to transfer rotational motion into linear motion

Basic Machine Elements

- **Bushing:** acts similarly to bearings; 2 types available: *ball bushing* and *oil-free bushing*; ball bushings are used in splines and have ball bearings present; oil-free bushings have imbedded lubricants instead of ball bearings

- **Snap Ring:** or *Retaining Rings* are usually used to prevent movement of bearings; 2 types: *C-ring* and *E-ring*; E-rings are usually used for smaller diameter shafts





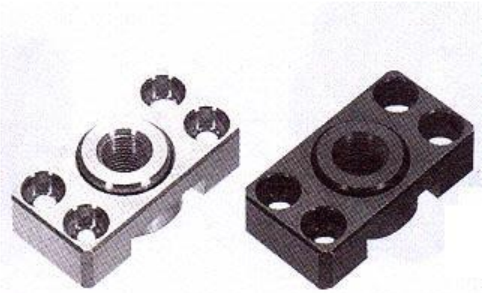
Basic Machine Elements

- **Washer:** helps in fixing of fasteners; 2 types: *flat washer* and *spring washer*; flat washers are used in slots while spring washers are used in environments where vibration is present

- **O-ring:** used to prevent leakage

Basic Machine Elements

- **Floating Joint:** usually used at the ends of cylinder rods to compensate for minor misalignment (eccentricity, deflection angle… etc.)



- **Power Lock:** used as a lock for shafts by method of gripping

- **Locating Pin:** for precise positioning of critical parts; prevents unnecessary movement; made of materials with high hardness

- **Shim:** used mainly as precise spacer, to take care of small adjustments necessary

Thank You!